Exercises

Logic Programming

https://moodle.aau.at/course/view.php?id=47556

University of Klagenfurt — Winter semester 2026

Homework 1 (Correction: 28 October 2026)

Exercise 1.1

Determine the answer set of the following positive logic programs:

1.1–a

```
a :- b, d.
a :- c, e.
b :- d, m.
b :- e, f.
c :- d, e.
d.
e :- d.
f :- b, c.
g :- d, f.
g :- e, h.
h :- c, k.
k :- a, e.
```

1.1–b

```
conn(amsterdam,brussels).
conn(brussels,cologne).
conn(cologne,amsterdam).

% tour(X,Z) :- conn(X,Y), conn(Y,Z).
tour(amsterdam,cologne)  :- conn(amsterdam,brussels),
                            conn(brussels,cologne).
tour(brussels,amsterdam) :- conn(brussels,cologne),
                            conn(cologne,amsterdam).
tour(cologne,brussels)   :- conn(cologne,amsterdam),
                            conn(amsterdam,brussels).

% finish(Y) :- tour(X,Y), conn(X,amsterdam).
finish(cologne)   :- tour(amsterdam,cologne),
                     conn(amsterdam,amsterdam).
finish(amsterdam) :- tour(brussels,amsterdam),
                     conn(brussels,amsterdam).
finish(brussels)  :- tour(cologne,brussels),
                     conn(cologne,amsterdam).
```

Exercise 1.2

Determine the reducts for the sets of atoms given in comments, and decide which of the sets of atoms are answer sets for the following normal logic programs:

1.2–a

```
%*
X1 = {}
X2 = {a,c}
X3 = {a,b,e}
*%

a :- not d.
b :- not a.
b :- e, not d.
c :- a, not e.
c :- b, not e.
d :- not a, not e.
e :- a, d.
e :- b, not c.
```

1.2–b

```
%*
X1 = {a,c,d,f}
X2 = {b,c,d,e}
X3 = {b,e}
*%

a :- not b.
b :- not c.
b :- not f.
c :- a.
c :- d.
d :- c, e.
d :- not b.
e :- not a, not f.
f :- not e.
```

Exercise 1.3

Reconsider the two logic programs in the previous exercise and find all their answer sets. (HINT: Avoid considering all of the possible interpretations.)

Exercise 1.4

Determine the answer sets of the following normal logic programs:

1.4–a

```
canary(tweety).
penguin(tux).

% bird(X) :- canary(X).
bird(tweety) :- canary(tweety).
bird(tux)    :- canary(tux).
% bird(X) :- penguin(X).
bird(tweety) :- penguin(tweety).
bird(tux)    :- penguin(tux).

% walks(X) :- penguin(X).
walks(tweety) :- penguin(tweety).
walks(tux)    :- penguin(tux).
% walks(X) :- bird(X), not flies(X).
walks(tweety) :- bird(tweety), not flies(tweety).
walks(tux)    :- bird(tux), not flies(tux).

% flies(X) :- bird(X), not walks(X).
flies(tweety) :- bird(tweety), not walks(tweety).
flies(tux)    :- bird(tux), not walks(tux).
```

1.4–b

```
num(1). num(2). num(3).

% lucky(X) :- num(X), not unlucky(X).
lucky(1) :- num(1), not unlucky(1).
lucky(2) :- num(2), not unlucky(2).
lucky(3) :- num(3), not unlucky(3).

% unlucky(X) :- num(X), not lucky(X).
unlucky(1) :- num(1), not lucky(1).
unlucky(2) :- num(2), not lucky(2).
unlucky(3) :- num(3), not lucky(3).

% p :- lucky(X), lucky(Y), X != Y, not p.
p :- lucky(1), lucky(2), not p.
p :- lucky(1), lucky(3), not p.
p :- lucky(2), lucky(3), not p.
```